CS 171: Discussion Section 1 (Sept 1)

1. Formal definitions

Provide formal definitions of (Gen, Enc, Dec) for the shift, substitution, and Vigenère ciphers. (You can use $\mathbb{Z}_n$ to denote the set of numbers $\{0, 1, \ldots, n-1\}$ and identify the letters of the English alphabet with $\mathbb{Z}_{26}$.)

Solution For set S, let $s \xleftarrow{\$} S$ denote selecting a uniformly random element s from S.

Shift Cipher.

1. Gen : Output $k \xleftarrow{\$} \mathbb{Z}_{26}$.
2. $\mathsf{Enc}(k, m = m_1 m_2 \cdots m_\ell)$: For $i \in \{1, 2, \ldots, \ell\}$, compute $c_i = m_i + k \mod 26$. Output $c = c_1 c_2 \cdots c_\ell$.
3. $\mathsf{Dec}(k, c = c_1 c_2 \cdots c_\ell)$: For $i \in \{1, 2, \ldots, \ell\}$, compute $m_i = c_i - k \mod 26$. Output $m = m_1 m_2 \cdots m_\ell$.

Substitution Cipher.

1. Gen : Let K indicate the set of all permutations $\pi : \mathbb{Z}_{26} \to \mathbb{Z}_{26}$. Output $\pi \xleftarrow{\$} K$.
2. $\mathsf{Enc}(\pi, m = m_1 m_2 \cdots m_\ell)$: For $i \in \{1, 2, \ldots, \ell\}$, compute $c_i = \pi(m_i)$. Output $c = c_1 c_2 \cdots c_\ell$.
3. $\mathsf{Dec}(\pi, c = c_1 c_2 \cdots c_\ell)$: For $i \in \{1, 2, \ldots, \ell\}$, compute $m_i = \pi^{-1}(c_i)$. Output $m = m_1 m_2 \cdots m_\ell$.

Vigenère cipher. For integer $z \in \mathbb{Z}$, let $[z] = \{1, 2, \ldots, z\}$.

1. Gen : Let T denote the maximum length of the period. Select $t \xleftarrow{\$} [T]$. For $\tau \in [t]$, select $k_\tau \xleftarrow{\$} \mathbb{Z}_{26}$. Output $k = k_1 k_2 \cdots k_t$.
2. $\mathsf{Enc}(k = k_1 k_2 \cdots k_t, m = m_1 m_2 \cdots m_\ell)$: Let $m_{i,\tau}$ indicate the τ-th character within the i-th block of t characters in m, namely $m_{i,\tau} = m_{(i-1)\cdot t+\tau}$ for $i \in [\lceil \ell/t \rceil], \tau \in [t]$. For $i \in [\lceil \ell/t \rceil]$ and $\tau \in [t]$, compute $c_{i,\tau} = m_{i,\tau} + k_\tau \mod 26$. Output $c = c_{1,1} c_{1,2} \ldots c_{1,t} c_{2,1} \ldots c_{n,t}$.
3. $\mathsf{Dec}(k = k_1 k_2 \cdots k_t, c = c_1 c_2 \cdots c_\ell)$: Let $c_{i,\tau}$ indicate the τ-th character within the i-th block of t characters in c, namely $c_{i,\tau} = c_{(i-1)\cdot t+\tau}$ for $i \in [\lceil \ell/t \rceil], \tau \in [t]$. For each $i \in [\lceil \ell/t \rceil]$ and $\tau \in [t]$, compute $m_{i,\tau} = c_{i,\tau} - k_\tau \mod 26$. Output $m = m_{1,1} m_{1,2} \ldots m_{1,t} m_{2,1} \ldots m_{n,t}$.

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2. Exploiting Partial Information About the Message

Setting: Let's say that a user wants to send one of two values over unsecured channels. For example, these could correspond to `Attack by land` and `Attack by sea`, or `abcd` and `ehgj`, or even something else. They will encrypt their message before sending it using one of the encryption schemes above and hope that an adversary who can see the ciphertext cannot figure out which message they sent.

In more general terms:

1. Assume that an adversary knows that a user's message is one of two values, m_A or m_B.
2. Say the user encrypts their message, and the adversary sees the resulting ciphertext.
3. Can the adversary figure out which message, m_A or m_B, was encrypted?

The answer depends on the particular values of m_A and m_B.

Questions:

(a) Say the user encrypts their message with the shift cipher, and say the message is either $m_A =$ `abcd` or $m_B =$ `ehgj`. Show how the adversary can determine the user's message, or show that this is not possible.

(b) Now say the user encrypts their message with the substitution cipher. If the message is either $m_A =$ `abcd` or $m_B =$ `ehgj`, can the adversary learn the message? If so, show how the adversary can do so. If not, find different values for m_A and m_B such that the adversary can learn the message.

(c) Say the user encrypts their message with the Vigenère cipher, and say the message is either $m_A =$ `abcd` or $m_B =$ `ehgj`. Show how the adversary can determine the user's password, or explain why this is not possible. Consider Vigenère ciphers that use period 2, period 3, and period 4.

Solution

1. For the case of the shift cipher, if the message was $m_A =$ `abcd` then the corresponding ciphertext will be 4 consecutive letters, while if the message was $m_B =$ `ehgj` then the corresponding ciphertexts will consist of nonconsecutive letters. Thus if the ciphertext consists of consecutive letters we can conclude that the message was m_A, and otherwise that it was m_B.

2. For the case of subsitution cipher, it is impossible to distinguish between encryptions of `abcd` and `ehgj`. Notice that if f is the chosen permutation then the ciphertext in the first case will be $(f(0), f(1), f(2), f(3))$ and the ciphertext in the second case will be $(f(4), f(7), f(6), f(9))$. If f is a random permutation then the distribution of these two ciphertexts will be the same.

   Alternatively, we can make it possible for the adversary to learn the message by choosing $m_A =$ `abcd` and $m_B =$ `aaaa`. Using the substitution cipher, the encryption of `abcd` is four different characters, whereas the encryption of `aaaa` is a single character repeated four times.

3. For the Vigenère cipher,

   (a) If the period was 2, then it is not possible to distinguish since the distance between `a` and `c` (resp. `b` and `d`) and `e` and `g` (resp. `h` and `j`) are the same. Hence, the distribution of the ciphertexts in both cases will be the same. Formally, if k_1 and k_2 are the chosen keys, then the first ciphertext will be $(k_1 + 0, k_2 + 1, k_1 + 2, k_2 + 3)$ (where $+$ denotes addition modulo 26) and the second ciphertext will be $(k_1 + 4, k_2 + 7, k_1 + 6, k_2 + 9)$. When k_1 and k_2 are randomly chosen from $\mathbb{Z}_{26}$, these two distributions are the same.

   (b) If the period was 3, then it is possible to distinguish since the distance between `a` and `d` is 3 whereas the distance between `e` and `j` is 5.

   (c) If the period was 4, it is not possible to distinguish between the two cases. Formally, if $k_1, \ldots, k_4$ are the chosen keys then the first ciphertext will be $(k_1 + 0, k_2 + 1, k_3 + 2, k_4 + 3)$ and the second ciphertext will be $(k_1 + 4, k_2 + 7, k_3 + 6, k_4 + 9)$. When $k_1, \ldots, k_4$ are randomly chosen, the distribution of these two ciphertexts are the same.

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